

GATES

Constitution gate → check traceability

Backlog **D3**. Every named constitution gate literal (CM-*) should be greppable inside the exact script that satisfies it, so a reviewer can go from the gate name in the governance docs (`_analysis/OPENDESIGN-ADOPTION-PLAN.md`, constitution submodule) straight to the runnable check and its evidence — no guessing, no bluff.

The gate literals are now embedded as `GATE: echo` lines near the top **and** in the final PASS/FAIL summary of each satisfying script, so:

```
grep -rn "CM-HOST-RENDERED-UI-VISUAL-PROOF\|CM-EXPORTED-DOC-VISUALLY-VALIDATED\|CM-OPENDESIGN-UI-SYSTEM" _tests _tools
```

lands on the check itself, not only on the docs.

The `GATE: echo` lines are **additive logging only** — they change no exit codes and no behaviour. The checks pass/fail exactly as they did before.

Traceability table

Gate literal	§	Satisfied by (exact command)	Evidence produced
CM-HOST-RENDERED-UI-VISUAL-PROOF	§11.4.170	<code>bash _tests/visual/self-validate.sh</code> (also run via <code>bash _tests/run-harness-selfvalidation.sh</code>). Drives <code>_tests/visual/visual-oracle.js</code> over	Per-theme host-rendered PNGs + oracle verdict under <code>_tests/evidence/harness/visual/</code> (golden-good., <i>golden-bad.</i>). Self-validated (golden-good/

Gate literal	§	Satisfied by (exact command)	Evidence produced
CM-EXPORTED-DOC-VISUALLY-VALIDATED	§11.4.168	fixtures/good.html (expect PASS) and fixtures/bad.html (expect FAIL); exits 0 only if good=PASS and bad=FAIL. bash _tests/export/self-validate.sh (also run via bash _tests/run-harness-selfvalidation.sh). Builds fixtures if missing, then drives _tests/export/validate-pdf.js over golden-good.pdf (expect PASS) and golden-bad.pdf (expect FAIL) across content + textual + full-visual (pdftotext/pdfimages/pdftoppm→OCR) layers; exits 0 only if good=PASS and bad=FAIL.	golden-bad), so the oracle is proven able to FAIL. Validator verdicts + extracted text/image/OCR artifacts under _tests/evidence/harness/export/. Self-validated (golden-good/golden-bad).
PORTFOLIO-DATA-INTEGRITY (sibling gate; not a CM-* literal)	§1.1 (anti-bluff)	bash _tools/portfolio/self-validate.sh. Drives _tools/portfolio/validate.mjs over selfvalidate-fixtures/good.json (expect PASS) and bad.json (expect FAIL); exits 0 only if the validator is mutation-paired (good=PASS, bad=FAIL).	golden-good.verdict.json + golden-bad.verdict.json under _tests/evidence/harness/portfolio/. The adoption plan (line 169) lists this as “a portfolio-data-integrity gate” alongside the three CM-* gates;

Gate literal	§	Satisfied by (exact command)	Evidence produced
			it has no CM-* literal of its own.
CM-OPENDESIGN-UI-SYSTEM	§11.4.162	No single self-validate exists. See the partial-evidence section below.	See below.

CM-OPENDESIGN-UI-SYSTEM — honest partial evidence (§11.4.6, no bluff)

There is **no** self-validated (golden-good/golden-bad) analyzer that satisfies CM-OPENDESIGN-UI-SYSTEM end-to-end. Do **not** treat this gate as green. What currently exists is a set of real artifacts that each evidence *part* of the “design-token / no-ad-hoc-CSS, both themes, AA contrast” intent:

- **Token-driven diagram rendering** — node `_tests/tools/diagram-brand-accent-proof.js` loads real generated product pages in a headless browser, reads the **computed** SVG stroke and the resolved `--od-accent` root token per `{light,dark}`, and asserts diagrams resolve to the correct brand token (company $\neq$ personal crimson). This proves colour comes from a resolved design token at render time, not an ad-hoc literal. Evidence: PNGs + `verdict.json` under `_tests/evidence/diagrams/brand-accent-proof/`.
- **Rendered-pixel AA contrast** — `_tests/tests/vasic-digital-ally.spec.js` and `_tests/tests/milosvasic-ru-ally.spec.js` (axe-core) keep the color-contrast (WCAG “serious”) rule enforced on real pages in the browser, evidencing that the token palette meets AA in the rendered UI (§11.4.107).
- **No-hardcoding regression** — `_tests/tests/v171-hardcoding.spec.js` guards against hardcoded/leaked content on rendered pages (content-level, not token-level).

Residual coverage gaps (do NOT claim these are covered)

- **No self-validated analyzer** for the gate itself (no golden-good/golden-bad proving the token check can FAIL) — unlike §11.4.168/.170.
- **No lint proving ZERO non-token colour/space literals** across CSS/inline styles. The adoption plan (line 100) demands “zero non-token color/space literals remain (lint)”; that lint is **not yet implemented**.

- **OpenDesign is not yet consumed as an external dependency.** §11.4.162 mandates OpenDesign as *the* token system consumed as a dependency, with tokens “never in CSS custom properties or inline styles.” The sites currently use bespoke `--od-*` CSS custom properties, so this core clause is **not satisfied** — the evidence above proves *token-driven* rendering, not *OpenDesign-sourced* tokens.

Until a self-validated token analyzer **and** the zero-literal lint exist, and the sites consume OpenDesign as a dependency, CM-OPENDESIGN-UI-SYSTEM is **partial / provisional**, not green.

Other CM-* literals in the docs (out of scope for D3)

These appear in the governance docs but are not wired by the three self-validate scripts above and are **not** claimed here: CM-NO-LOCAL-RUNTIME, CM-SCRIPT-DOCS-SYNC, CM-TEST-CATALOG-FRESH, and the CM-COVENANT-114-* -PROPAGATION family. They are listed for completeness so this file is not mistaken for a full gate registry.

How to re-verify

```
# §11.4.170 + §11.4.168 – must end with: OVERALL RESULT: PASS
bash _tests/run-harness-selfvalidation.sh 2>&1 | tail -4

# portfolio-data-integrity – must be rc=0
bash _tools/portfolio/self-validate.sh; echo rc=$?

# gate literals are now greppable inside the checks
grep -rn "CM-HOST-RENDERED-UI-VISUAL-PROOF\|CM-EXPORTED-DOC-
VISUALLY-VALIDATED\|CM-OPENDESIGN-UI-SYSTEM" _tests _tools
```